The Principal Burden and the Long-Haul Invariant

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November 3, 2026

Abstract

Platform logistics firms in Kenya and comparable African markets have failed repeatedly across the last mile (Sendy), fresh produce distribution (Twiga), and long-haul freight (Lori Systems). This paper isolates one mechanism common to all three: the **principal burden** Φ, the cost of financing the informality of the supplier base on the firm's own balance sheet. We define $\Phi = v(1-\kappa) + r\Delta t$, decomposing it into an input-credit gap ($v(1-\kappa)$) and a working-capital gap ($r\Delta t$), and show how it enters the viability invariant as the term ΦC_p. We state the long-haul form of the invariant, formalise the Long-Haul Proposition, and calibrate the model on a worked long-haul case in which $\Phi C_p \approx$ KSh 7,065. A reconciliation of Sendy, Twiga, and Lori Systems shows that each failure is the same integrand integrated over a different domain.

1 The Principal Burden and the Long-Haul Invariant

When the firm buys capacity and resells it—*whichever contractual label it uses*—a second cost enters that does not depend on addressability but does depend on the firm's position between an informal supplier base and a formal customer base.

Definition 1 (Principal burden)**.** *Let C_p be partner capacity cost per trip, κ the share of VAT-registered partners, v the VAT rate, r the annualised cost of capital, and Δt the payment gap in years. Define:*

$$\Phi = v(1-\kappa) + r\Delta t \tag{1}$$

so that the principal burden per trip is ΦC_p.

The two terms have distinct mechanisms:

- $v(1-\kappa)$ is the **input-credit gap**: VAT owed on gross without recovery on inputs, arising because the informal supplier base is not VAT-registered. It arrives retroactively in a *reclassification* case and structurally in an *embedded-principal* case.
- $r\Delta t$ is the **working-capital gap**: the cost of paying the supplier before the customer pays.

Both are the cost of **financing the informality of the supplier base on the firm's own balance sheet**.

The Long-Haul Invariant:

$$\boxed{\bar{n}\, p(a)\, A(a) > c_{\text{mat}} + \frac{w_{\text{driver}}}{R(a)} + (1-p(a))c_r + C_p\big[v(1-\kappa) + r\Delta t\big]} \tag{2}$$

For a long-haul trip, $\bar{n}$ is large and endpoint failures are few but expensive. The principal burden is proportional to C_p, which scales with the value of the goods moved. **Long-haul does not escape the address problem; it multiplies the principal burden per endpoint.**

2 The Long-Haul Proposition

Proposition 1 (Long-Haul Proposition). *For long-haul trips, $N_{trip} \to 2$ (origin and destination), so the amortisation channel collapses: each endpoint's failure is borne in full, not divided across many. Simultaneously the principal burden ΦC_p enters the invariant and scales with the value of the goods moved. Hence the long-haul invariant is dominated by endpoint addressability at a_{origin} and $a_{destination}$ multiplied by asset opportunity cost λ, and by the principal burden—not by fuel or distance, which enter linearly and predictably.*

3 Case E: A Worked Long-Haul Calibration

Case E — Long haul (illustrative). $\bar{n} = 60{,}000$, $C_p = 45{,}000$, $\kappa = 0.2$, $v = 0.16$, $r = 0.18$, $\Delta t = 0.16$ (60-day gap). Then $\Phi = 0.16 \times 0.8 + 0.18 \times 0.16 = 0.128 + 0.029 = 0.157$, and $\Phi C_p \approx 7{,}065$. A single endpoint failure at $a = 0.2$ with $\lambda = 300$ adds $\lambda(1-a)/a = 1{,}200$ to service cost—and the truck waiting at that gate is not moving goods worth $\bar{n}$. Both the address term and the principal term are large enough to turn a viable trip negative.

4 Reconciliation: Sendy, Twiga, Lori Systems

Reconciliation.

- **Sendy** operated at low $\bar{n}$, low a, with fulfilment expansion raising c_0. Its principal burden arrived *retroactively* as a $v(1-\kappa)$ shock on the entire book at once—an acute presentation of Φ.
- **Twiga** owned c_0 (warehouses, fleet, farming) while operating in fresh produce with low $\bar{n}$ and high variance, requiring N_{trip} above what informal addressing could supply.
- **Lori Systems** operated at high $\bar{n}$ and low a at both endpoints, with $N_{\text{trip}} = 2$ so the amortisation channel collapsed. Its principal burden arrived *structurally* and chronically as $r\Delta t$ on a 30–90 day gap. A 96% valuation write-down priced the variance that addressability could not remove.

5 The Same Integrand, Different Domain

The same integrand, different domain. The address term is $\lambda \cdot (1-a)/a$ at every scale. Only λ changes: a boda-boda idling at a gate costs shillings per minute; a 40-tonne truck idling at an unlocatable yard costs orders of magnitude more, and the truck's fixed cost per day is enormous relative to the number of trips it can complete. **Sendy and Lori are the same failure integrated over different domains—one over many small endpoints, the other over few large ones—and neither owned the address layer that would have made the endpoint calculable and the principal burden closeable.**

6 Conclusion

The principal burden $\Phi = v(1-\kappa) + r\Delta t$ is not an accounting detail but a structural cost of sitting between an informal supplier base and a formal customer base. In the long haul, where $\bar{n}$ is large and the amortisation channel collapses, the burden is proportional to C_p and scales with the value of the goods moved. Long-haul does not escape the address problem; it multiplies the principal burden per endpoint—and the same failure integrates over few large endpoints instead of many small ones.